



WELCOME TO

SITH

**Hardware:
Price Increase**

Tuesday 1/20

2020

2018



Debrief

Main goal of this meeting:

- **Discuss the extreme RAM and SSD price increases**
- **Discuss the past chip shortages**





AI Chatbots

As of the past 2 years, AI chatbots and AI generation tools like Chat GPT, Sora AI, Gemini, Copilot, Claude, etc have become a regular tool like Google.

AI is an extremely demanding tool requiring enormous data centers to develop, train, and run these tools.



2020

2018



AI Data Centers & RAM

- Like any computer, servers running LLMs (large language models e.g. ChatGPT) require RAM, and LOTS.
- AI companies like OpenAI, Google, etc have purchased roughly 25-50% of all UNPRODUCED RAM from manufacturers. This RAM is scheduled to go into datacenters that have NOT yet been built, with money that is NOT yet available.
- RAM companies like Micron, popular for their Crucial line of products left the consumer space to sell directly to AI companies, leaving us with even less options.



AI Data Centers & SSDs

- The same main manufacturers produce SSDs for consumers and data centers
- AI data centers need to store ALL user generated content, such as brainrot videos, chats, and even things you cant see like the AIs “memory”, training, programming and more.
- Significant shifts in supply from consumer SSDs to enterprise grade SSDs as the demand is exponentially higher.

2020

2018



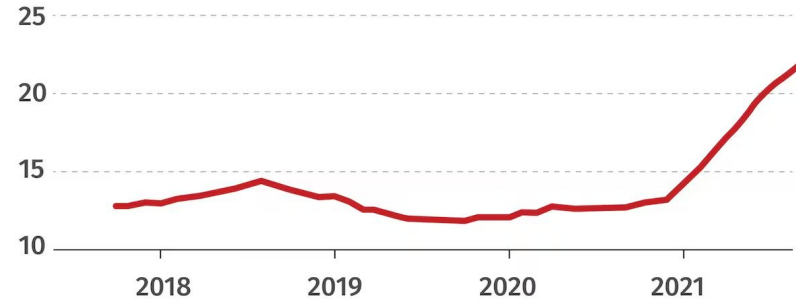
2020-23 Chip Shortages

- **Significant demand of chips due to Covid-19 and everyone being stuck at home purchasing computers for school, work, and gaming led to a several-year long shortage, increasing prices of (mainly) GPUs**

Long wait for microchips

Factories are facing delays of up to 22 weeks in getting microchips

No. of weeks



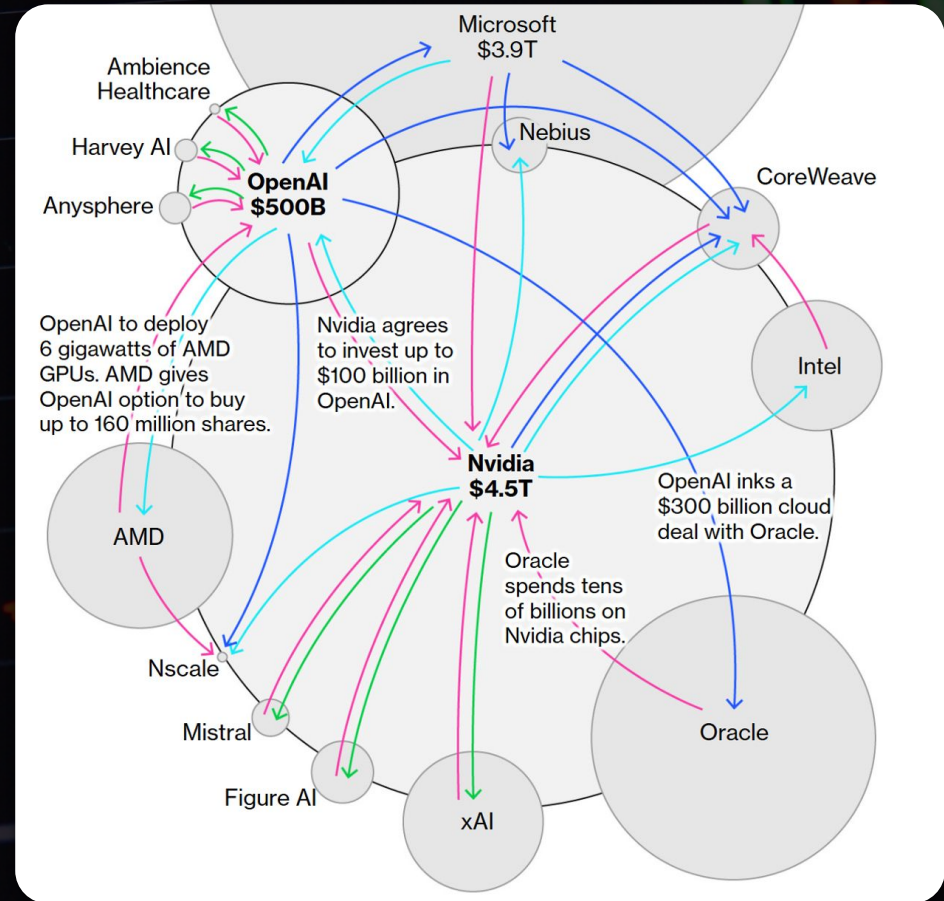
The 'AI Circle'

How Nvidia and OpenAI Fuel the AI Money Machine

— Hardware or Software
 — Investment
 — Services
 — Venture Capital
 Circles sized by market value

Referring back to the “unproduced” and “not built” emphasis’ – AI companies are all passing money around in a “self-reinforcing loop.”

Some doubt the economical principles supporting this ‘bubble,’ but others claim it is entirely ok.





The Keystones

All chips come from photolithography machines developed by ASML, who then sells these several-hundred million dollar machines to companies to make CPUs, GPUs, RAM, etc. The most significant being TSMC. TSMC takes the machines and actually fabricates the chips for RAM, SSDs, CPUs, etc for companies like AMD and Nvidia.

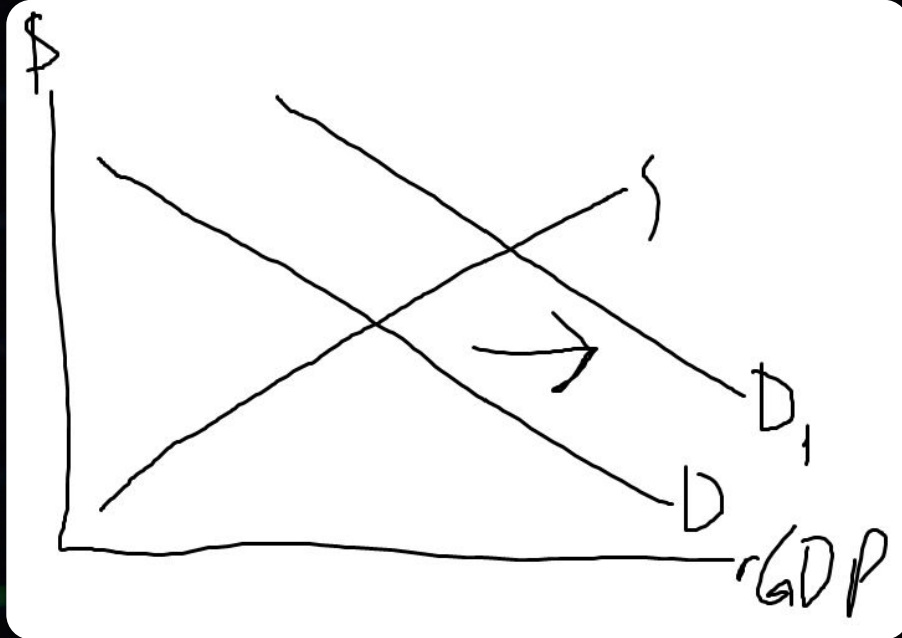
Dozens, if not hundreds of companies relying on only 2 companies to produce CPUs, GPUs, RAM, and dozens of other crucial parts of computers is ground for a bad day, and exactly the cause of the chip shortage in 2020-23.



ASML

2020

It all comes down to this...



- As demand increases, prices increase





Next Meeting

1/27

Ground plan:

- TBD



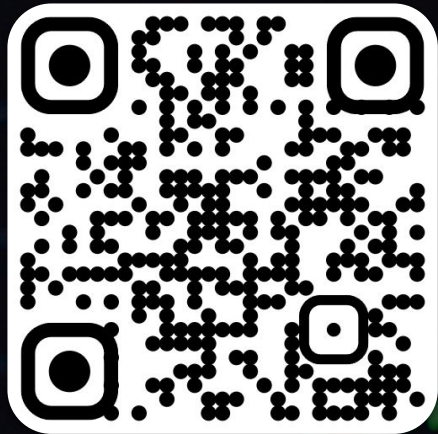
2020

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Info

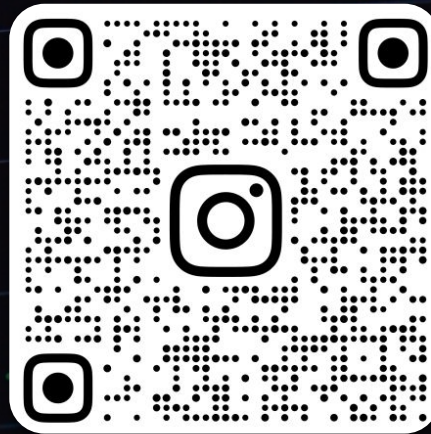
Discord



Website



Instagram



**Hall Pass
Demo**



Note: Find all past presentations and other media on our website!